



# From Accounts to Alerts: EDA and Logistic Regression for Early Corporate Risk Prediction

Yun-Hsuan Chang, Chia-Yen Chang, Huei-Wen Teng



# Outline

1. Motivation
2. Data
3. Methodology



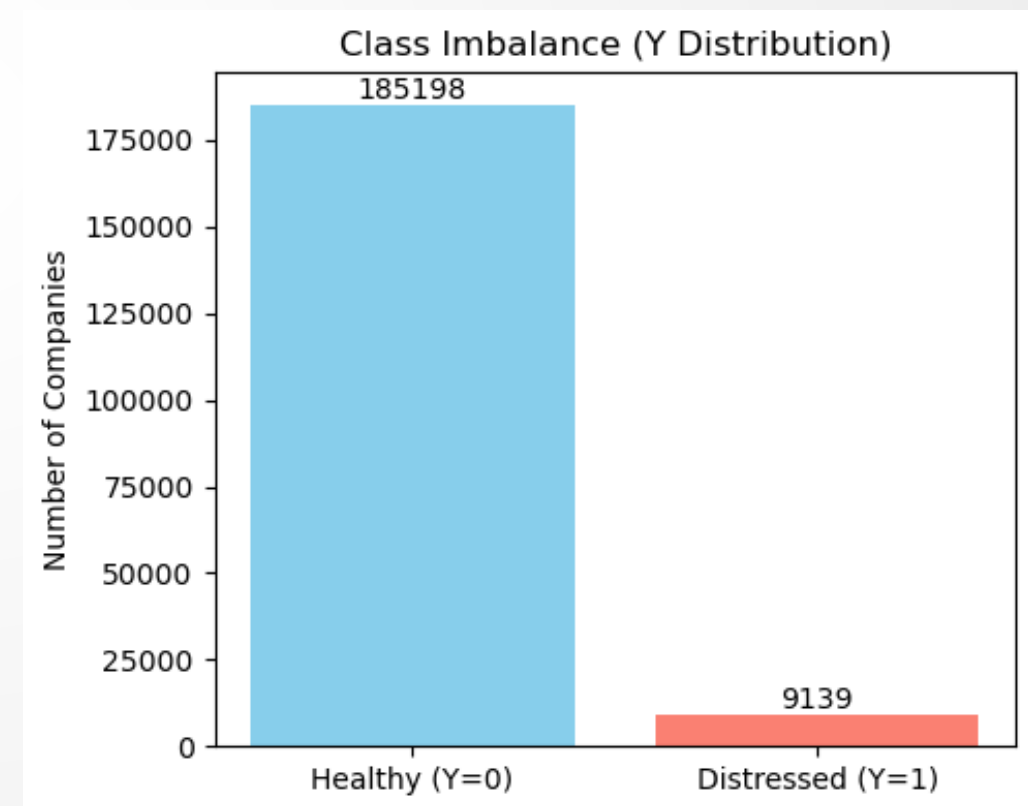
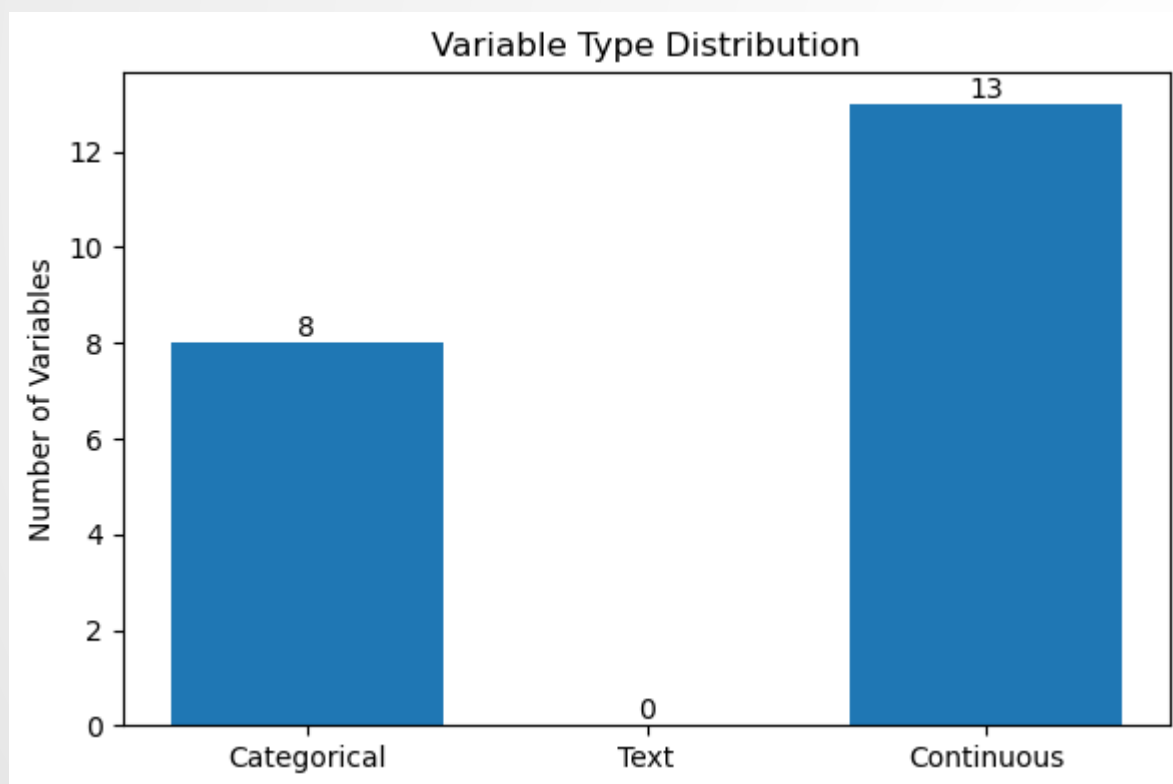
# Motivation

- Early Warning Systems
- Transparency and Interpretability
- Publicly Available Data



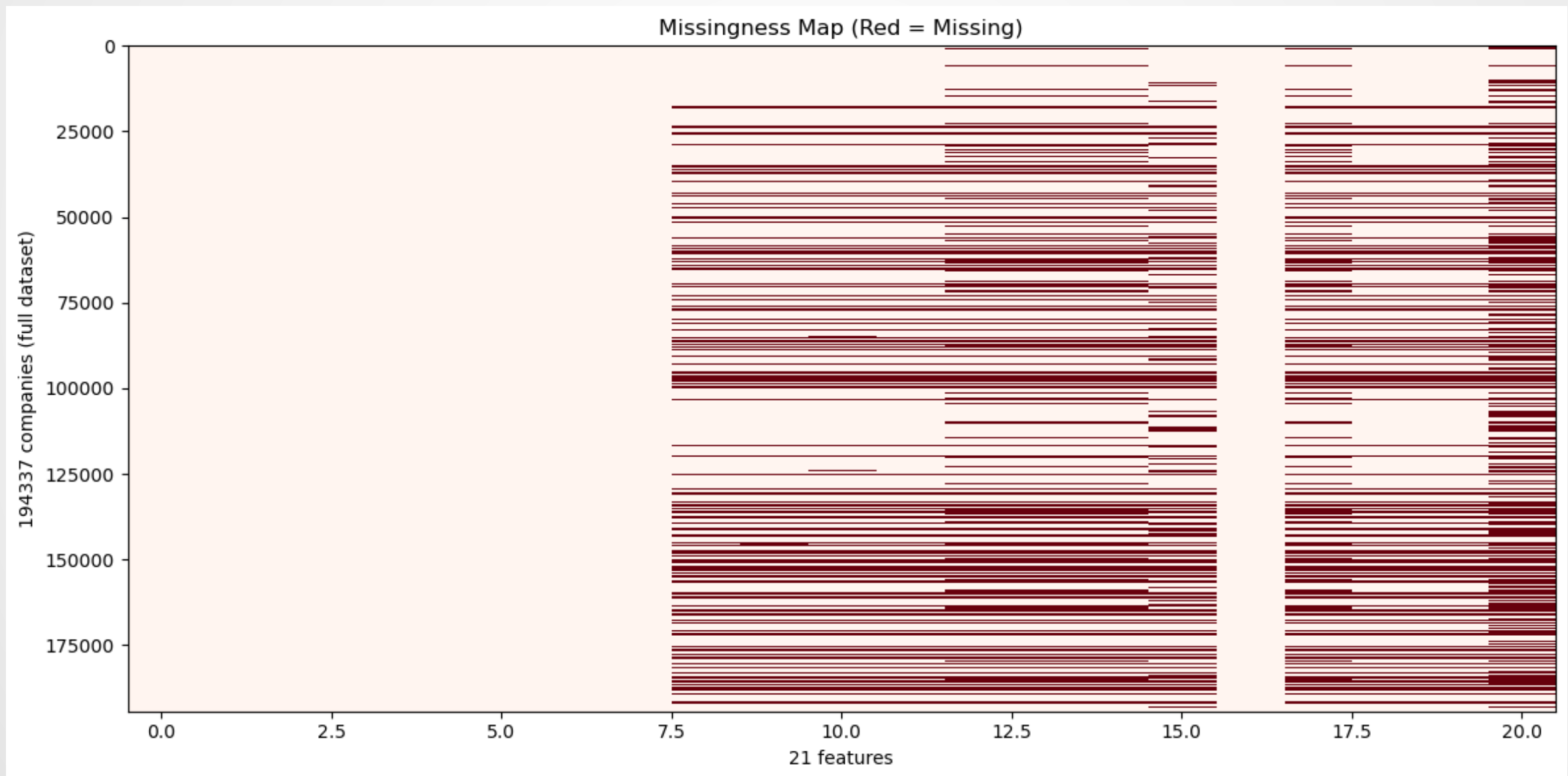
# Data

- companies: 23033
- features: 21
- Class imbalance (Y=1 ratio): 0.0470



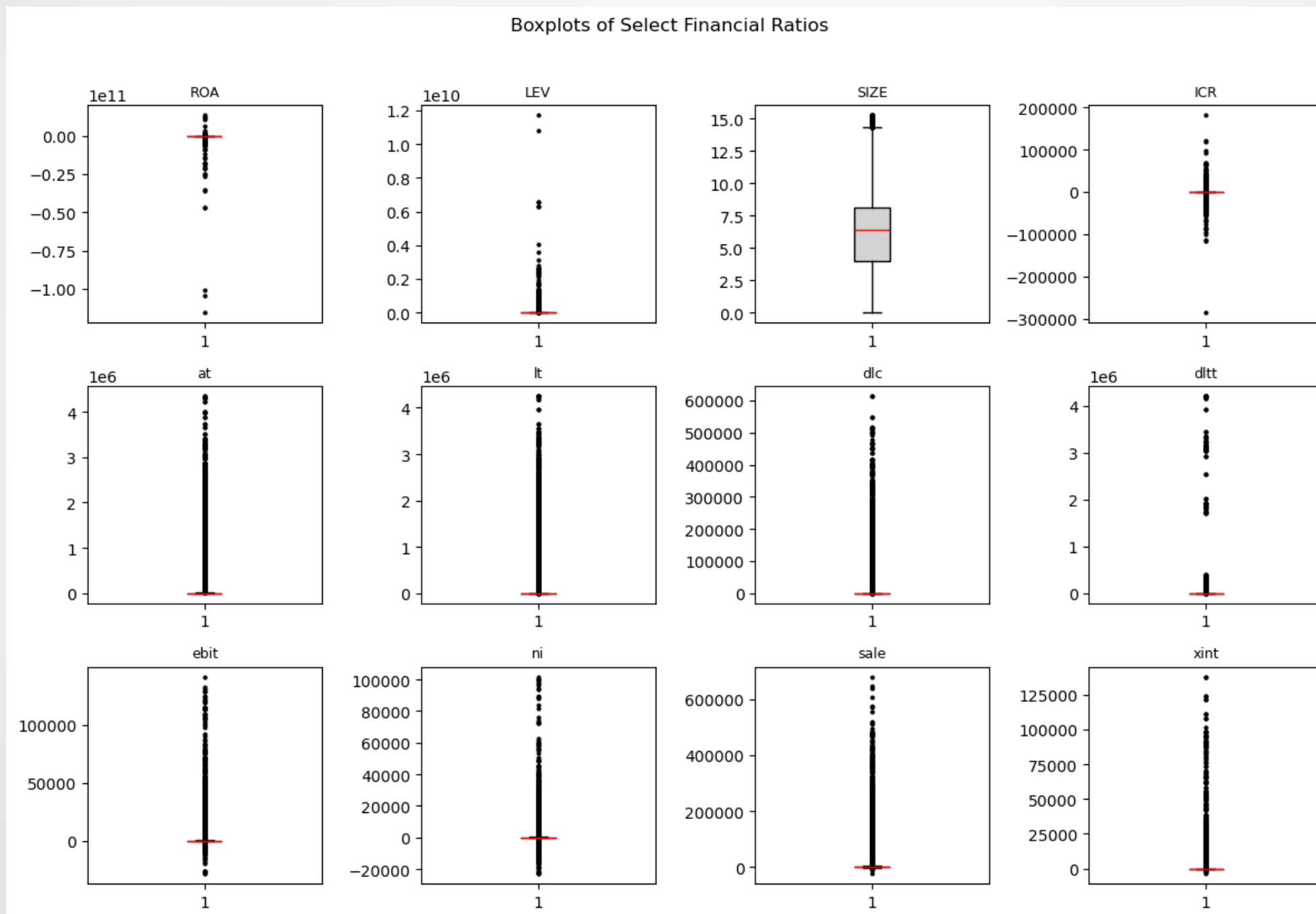
# Data

- Average missing rate across all features: 18.17%
- Impute: SIC2×year median (train-only); else overall train median



# Data

- heavy-tailed and asymmetric distributions



## Data

- Most variables are highly skewed
- Several variables exhibit extreme kurtosis ( $>10$ )
- apply winsorization and log-transform before model estimation

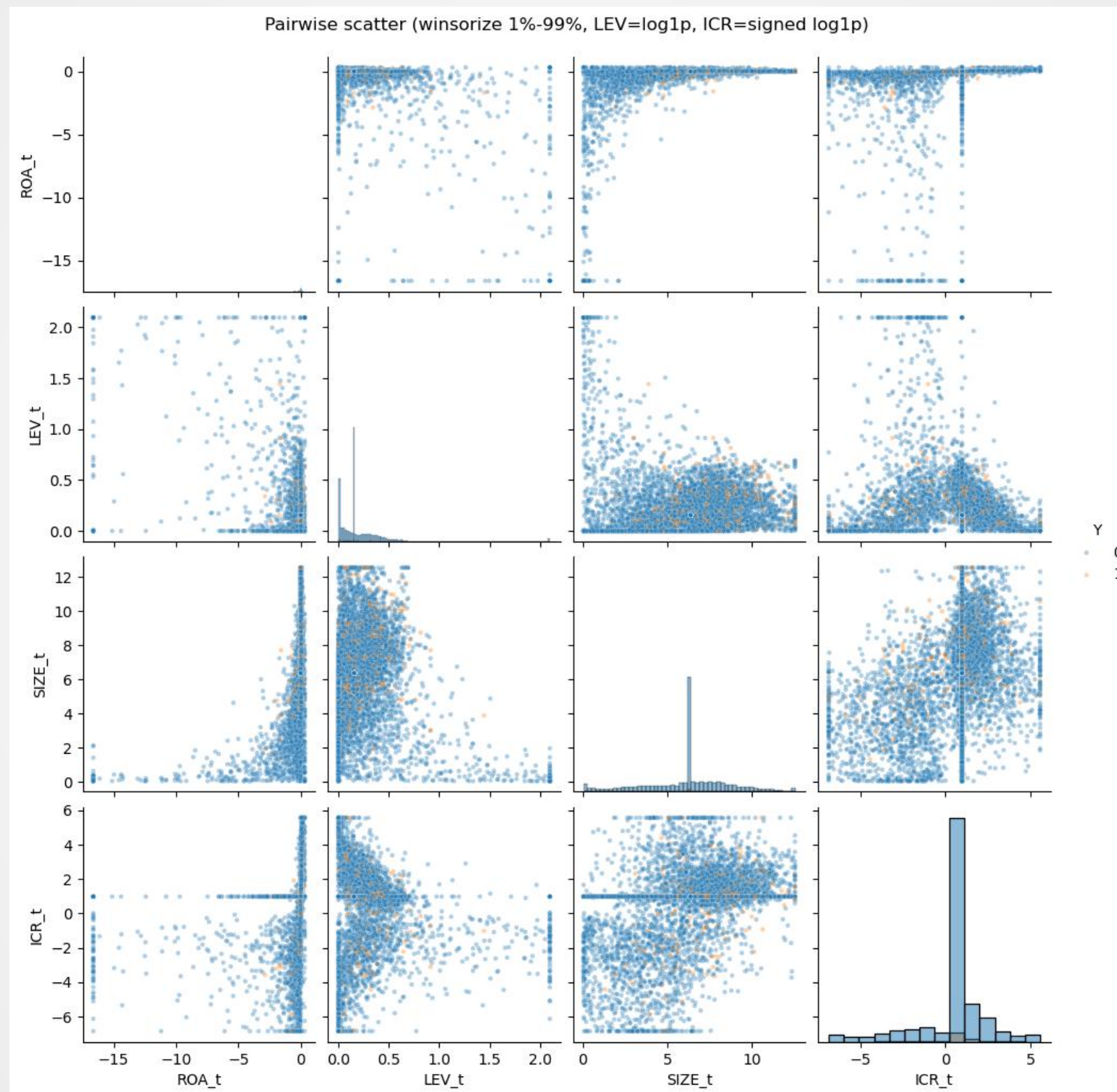
### [Descriptive Summary of Selected Numerical Features]

|      | count    | mean        | std          | min           | 25%   | 50%    | 75%     | max          | skewness | kurtosis |
|------|----------|-------------|--------------|---------------|-------|--------|---------|--------------|----------|----------|
| ROA  | 126632.0 | -6869530.06 | 6.125031e+08 | -1.154960e+11 | -0.19 | 0.00   | 0.05    | 1.350200e+10 | -140.97  | 23346.64 |
| LEV  | 143774.0 | 1144596.71  | 6.431927e+07 | 0.000000e+00  | 0.03  | 0.17   | 0.41    | 1.172300e+10 | 113.36   | 16453.89 |
| SIZE | 143956.0 | 6.12        | 2.960000e+00 | 0.000000e+00  | 4.02  | 6.37   | 8.14    | 1.529000e+01 | -0.08    | -0.35    |
| ICR  | 94503.0  | -54.03      | 2.173370e+03 | -2.855470e+05 | -4.85 | 1.67   | 6.85    | 1.817916e+05 | -19.70   | 4470.59  |
| at   | 143956.0 | 20730.17    | 1.526165e+05 | 0.000000e+00  | 54.88 | 582.78 | 3439.39 | 4.349731e+06 | 14.35    | 247.71   |
| lt   | 143773.0 | 17758.46    | 1.431946e+05 | -1.000000e-01 | 17.28 | 315.03 | 2244.00 | 4.255074e+06 | 14.88    | 266.24   |
| dlc  | 142956.0 | 1433.99     | 1.560927e+04 | -5.600000e+01 | 0.00  | 4.41   | 62.90   | 6.142374e+05 | 19.34    | 449.17   |
| dltt | 143605.0 | 3902.35     | 6.283327e+04 | -2.000000e-02 | 0.04  | 34.15  | 669.61  | 4.216909e+06 | 49.55    | 2719.41  |
| ebit | 126630.0 | 528.02      | 3.263470e+03 | -2.838700e+04 | -4.10 | 6.86   | 143.73  | 1.412900e+05 | 18.27    | 499.06   |
| ni   | 126636.0 | 271.61      | 1.971740e+03 | -2.311900e+04 | -9.21 | 1.17   | 68.64   | 1.018320e+05 | 20.72    | 746.89   |
| sale | 126639.0 | 3703.27     | 1.790896e+04 | -2.495468e+04 | 7.12  | 141.59 | 1219.69 | 6.809850e+05 | 13.86    | 283.16   |
| xint | 127418.0 | 211.92      | 2.356540e+03 | -3.775000e+03 | 0.16  | 4.70   | 51.11   | 1.378610e+05 | 31.93    | 1255.55  |





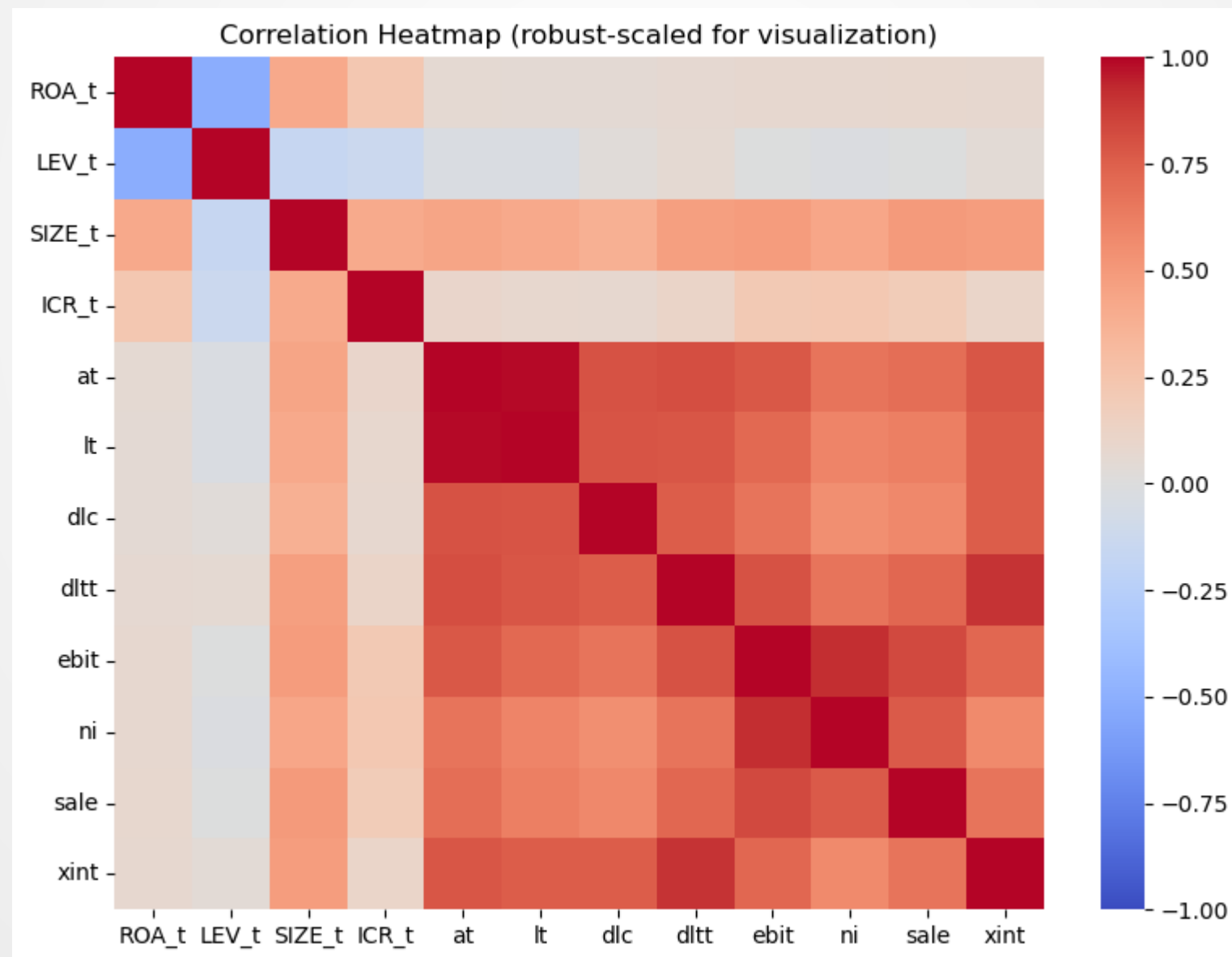
# Data





## Data

- Positive correlations among size-related variables (*at*, *lt*, *sale*, *ni*)
- ROA inversely related to LEV (higher leverage, lower profitability)
- No major multicollinearity after data transformation



## Overview

- USA annual financial statements (2019–2024) + SEC Form 25 delisting labels
- Label:  $Y=1$  if a Form 25 becomes effective within 400 days after the financial statement date
- Train: 2019–2023, Test: 2024

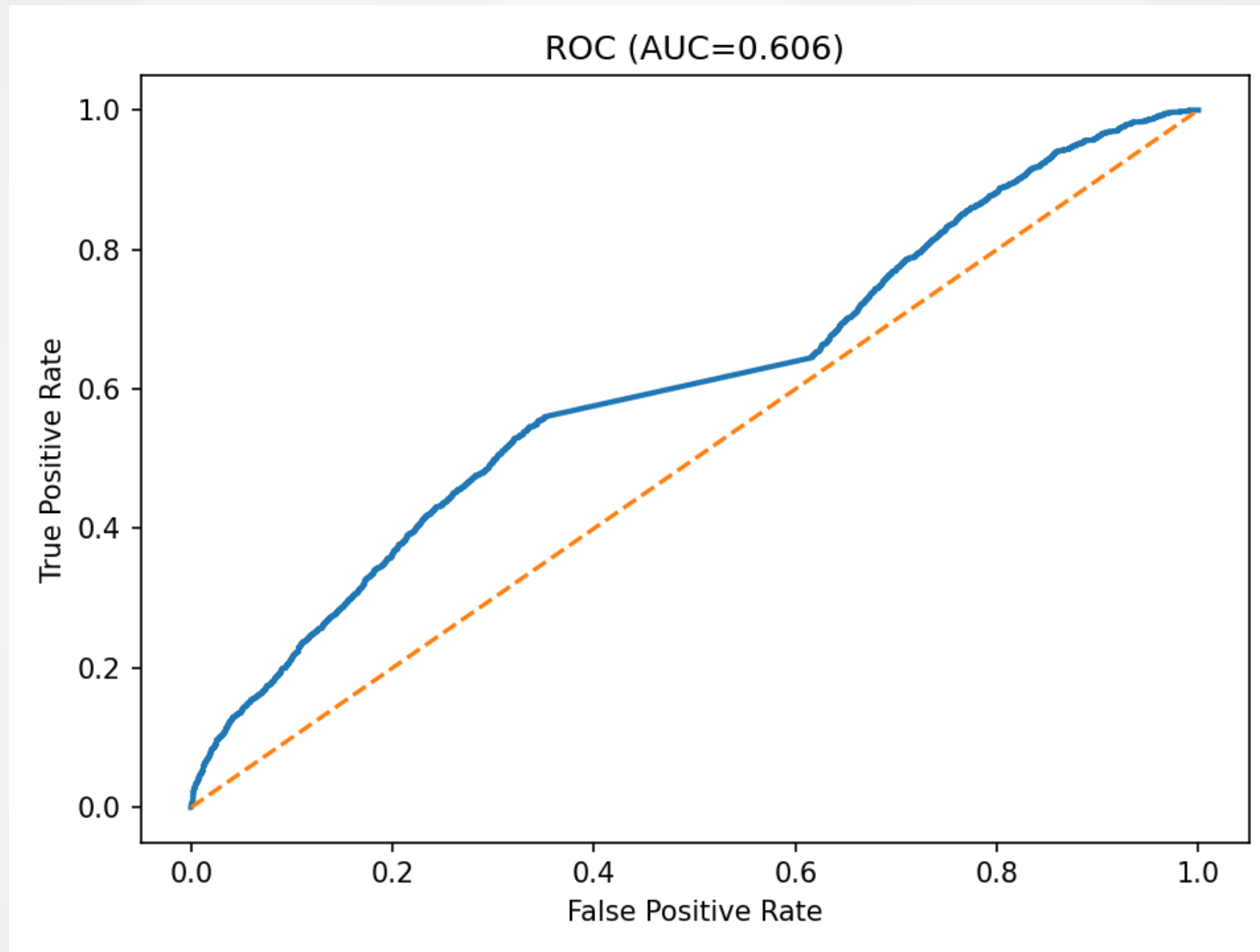


## Benchmark Model

- ▣ Penalized logistic regression (L2 / Ridge) as baseline
- ▣ Mitigates overfitting and stabilizes coefficients
- ▣ Addresses class imbalance with **class\_weight="balanced"**
- ▣ Evaluated by AUC, PR-AUC, Brier score, Calibration, and Youden's J



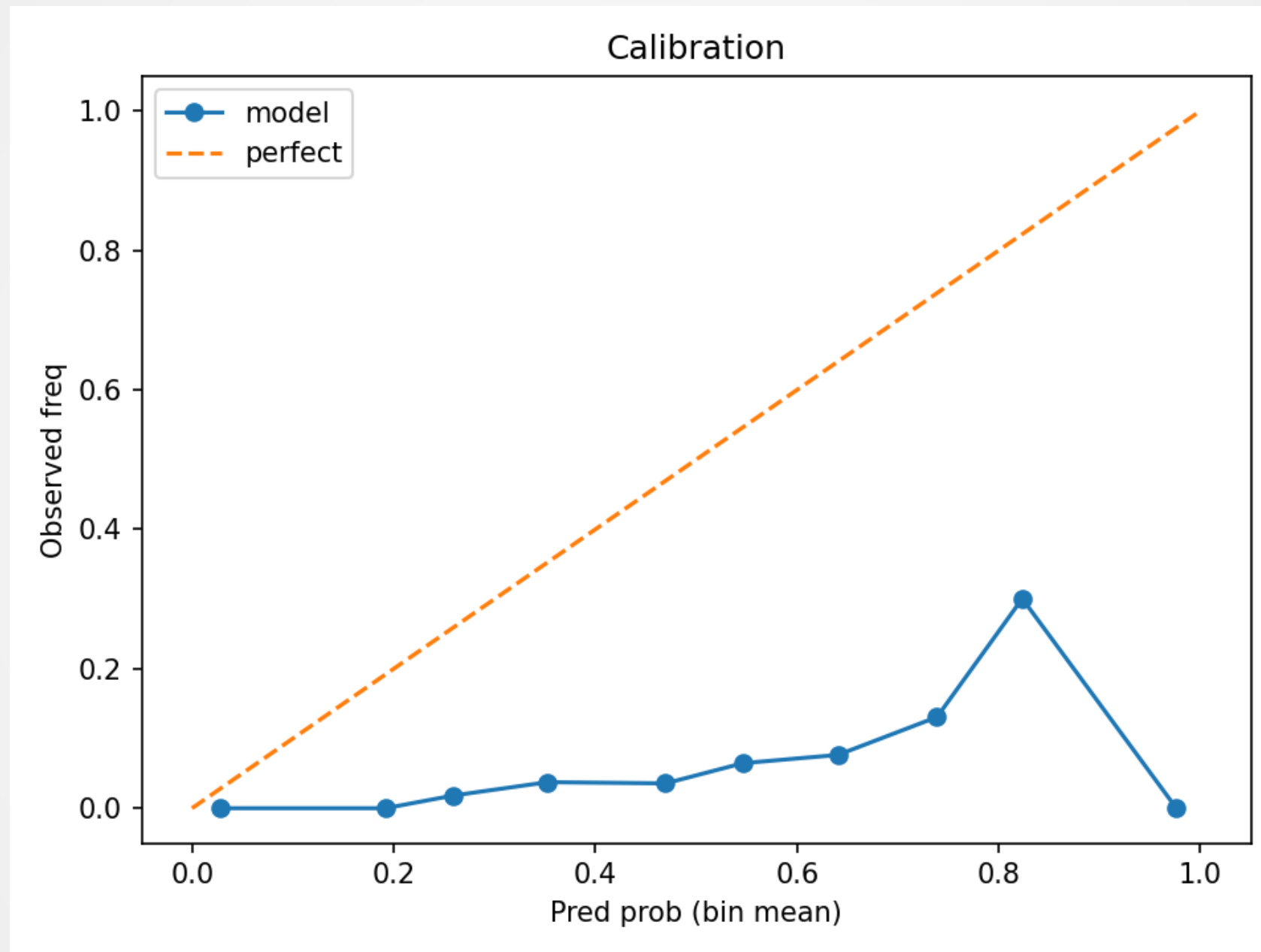
## Benchmark Model



Baseline L2-penalized logistic regression achieved AUC = 0.61, indicating moderate discriminatory power.



## Benchmark Model (Calibration Analysis)



The baseline logistic model tends to overestimate default risk, especially in mid-to-high probability ranges (0.3–0.8).



## Class Imbalance Handling

- ▣ **Baseline:** Cost-sensitive weighting (`class_weight='balanced'`)
  - Adjusts loss to account for rare delisting events.
- ▣ **Extension:** Synthetic Minority Oversampling Technique (**SMOTE**)
  - Generates synthetic samples of the minority class to improve recall





## Design of Research

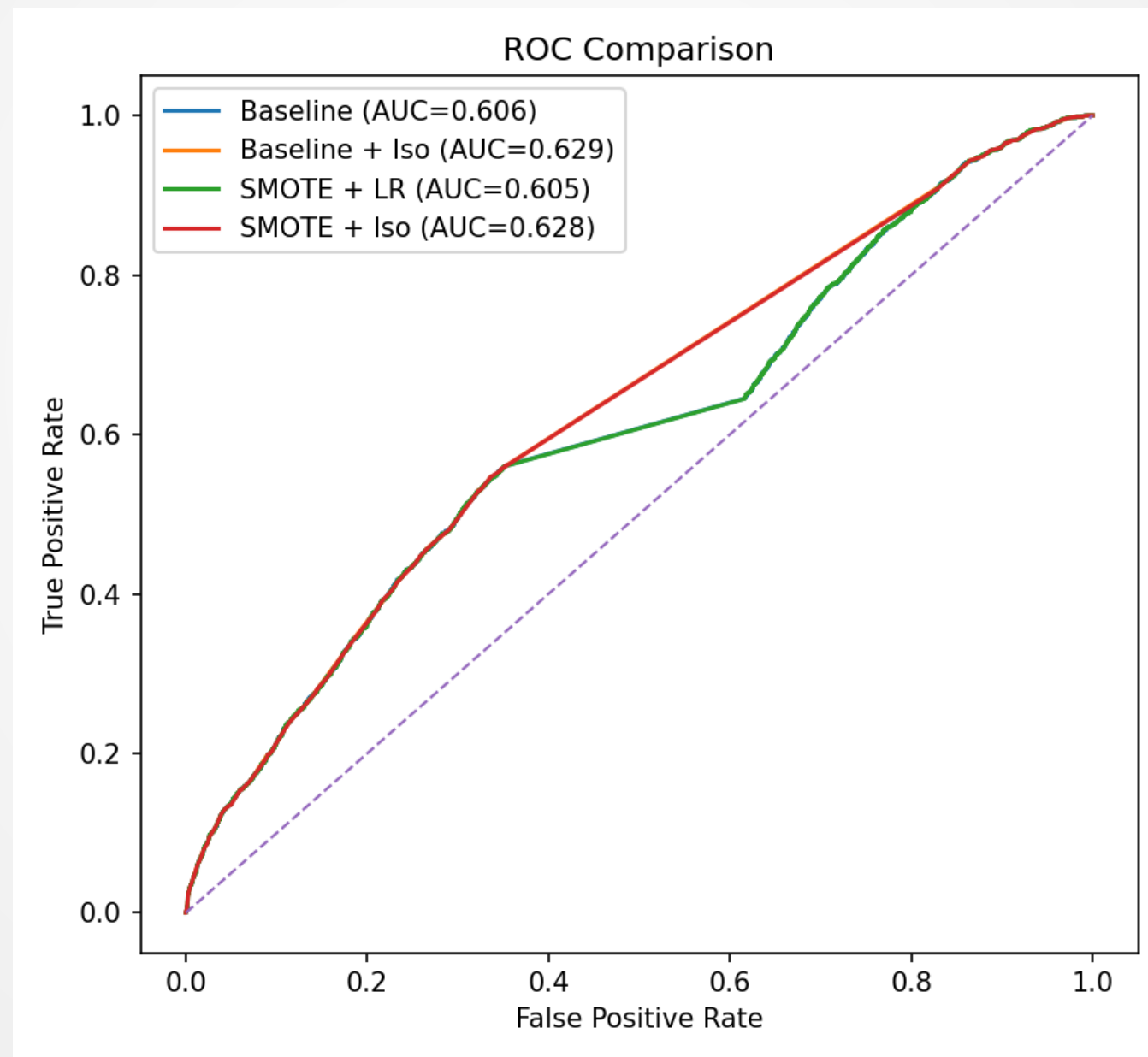
- ▣ EDA & Feature Engineering: clean, transform, and visualize financial ratios
- ▣ Benchmark Model: L2-regularized logistic regression with `class_weight="balanced"`
- ▣ Calibration Analysis: evaluate AUC, PR-AUC, Brier, and calibration curve
- ▣ Post-hoc Improvements:
  - *SMOTE* for class imbalance
  - *Platt scaling* and *Isotonic regression* for probability calibration
- ▣ Evaluation & Interpretation: discrimination (AUC, PR-AUC), calibration, and coefficient analysis



## Post-hoc Improvements

- Post-hoc improvements include **SMOTE** for imbalance and **Platt/Isotonic** for probability calibration.

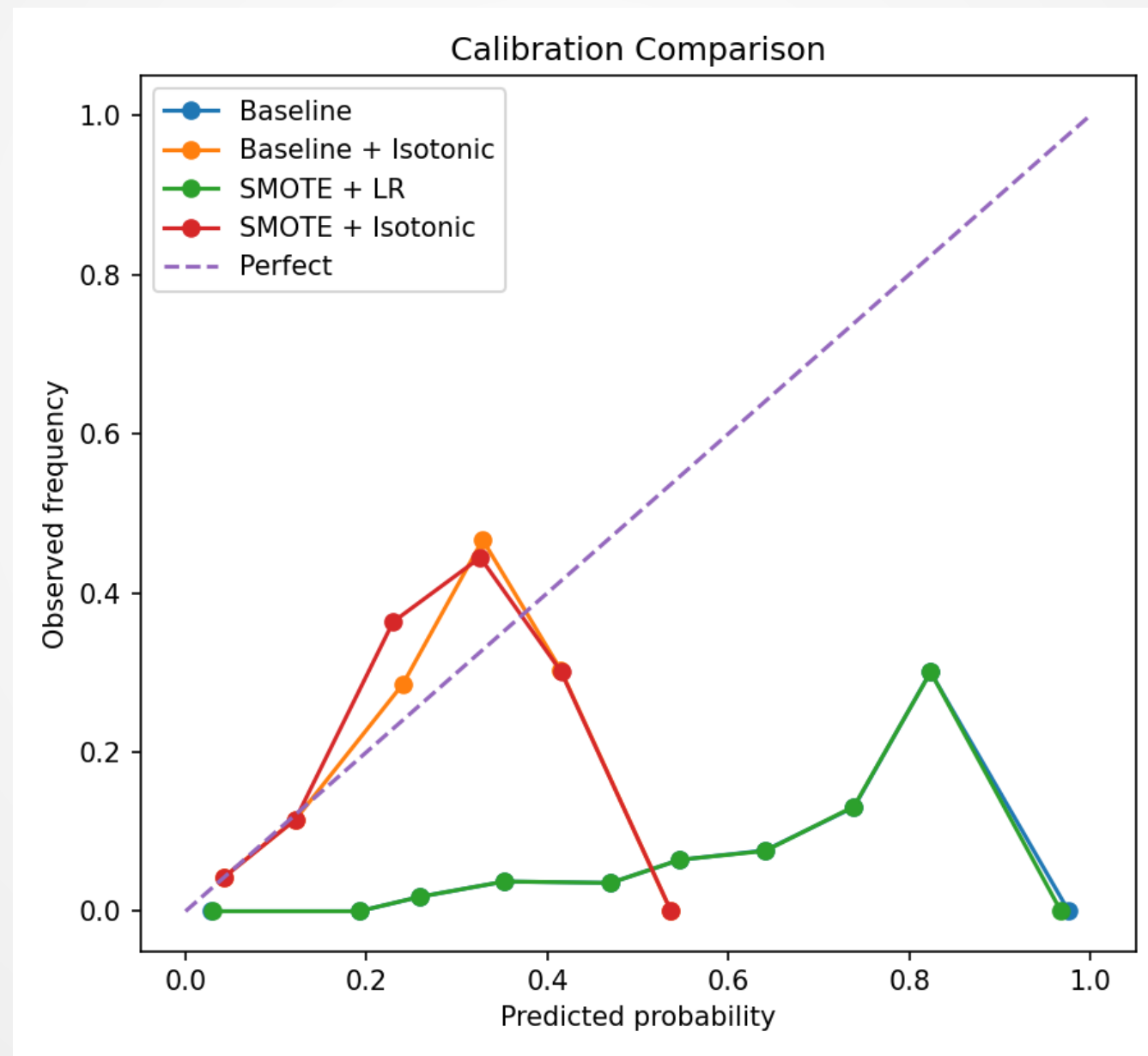
Calibration and resampling (SMOTE) slightly improve discrimination, as reflected by higher AUC.”



# Post-hoc Improvements

- Post-hoc improvements include **SMOTE** for imbalance and **Platt/Isotonic** for probability calibration.

Post-hoc calibration (Isotonic) improves probability reliability, especially under imbalanced data (SMOTE).



## Performance Measures

- Discrimination: AUC / ROC, H-measure, and Precision–Recall curves
- Calibration: Brier score and calibration plots
- Threshold-based metrics: accuracy, sensitivity, specificity, F1-score





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